

```

8   'Corn Stover': 1100,
9   'Paddy Straw': 1300,
10  'Coconut Shell': 1400,
11  'Groundnut Shell': 1200,
12  'Mustard Stalks': 1080,
13  'Other': 1000,
14 };
15
16 // Calculate carbon offset for diverted waste
17 export function calculateCarbonOffset(
18   wasteType: string, quantityMT: number
19 ): number {
20   const factor = CARBON_FACTORS[wasteType] || CARBON_FACTORS['
      Other'];
21   return factor * quantityMT;
22 }

```

Listing 8.5: Carbon Credit Service (carbon-service.ts)

The `CARBON_FACTORS` constant maps each waste type to its CO₂ equivalent savings in kilograms per metric tonne. The `calculateCarbonOffset` function provides a simple multiplication-based calculation with a fallback to the “Other” category for unrecognized waste types.

8.6 TypeScript Type Definitions

AgriTrace uses TypeScript interfaces and enums to enforce type safety across the entire codebase. The `types.ts` module defines all data models used by the application:

```

1  import { Timestamp } from 'firebase/firestore';
2
3  export enum UserRole {
4    FARMER = 'farmer',
5    AGENT = 'agent',
6    ADMIN = 'admin',
7  }
8
9  export enum WasteReportStatus {
10    PENDING_PICKUP = 'Pending Pickup',
11    IN_TRANSIT = 'In Transit',
12    DELIVERED = 'Delivered',
13    PROCESSING = 'Processing',
14    RECYCLED = 'Recycled',
15    REJECTED = 'Rejected',
16    COMPLETED = 'Completed',

```